NXT007: The Future of Hemophilia A Treatment

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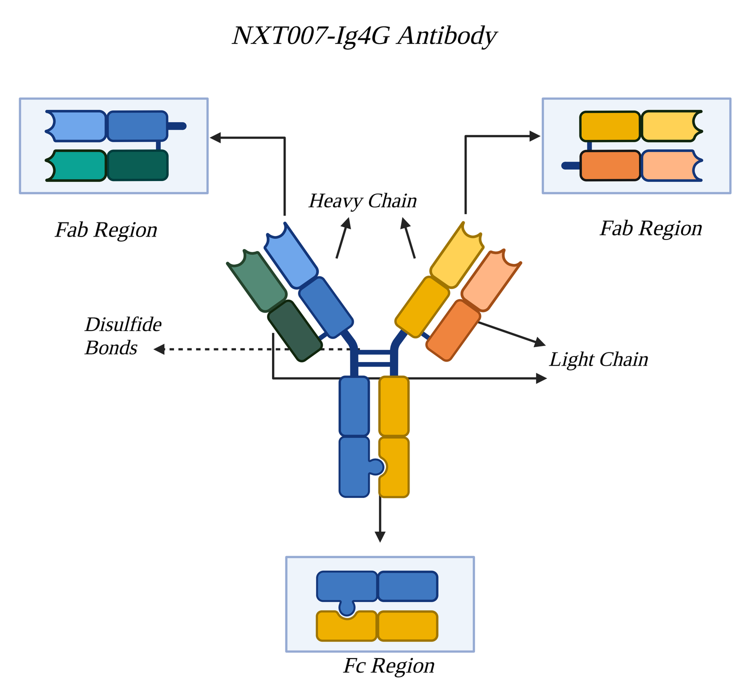
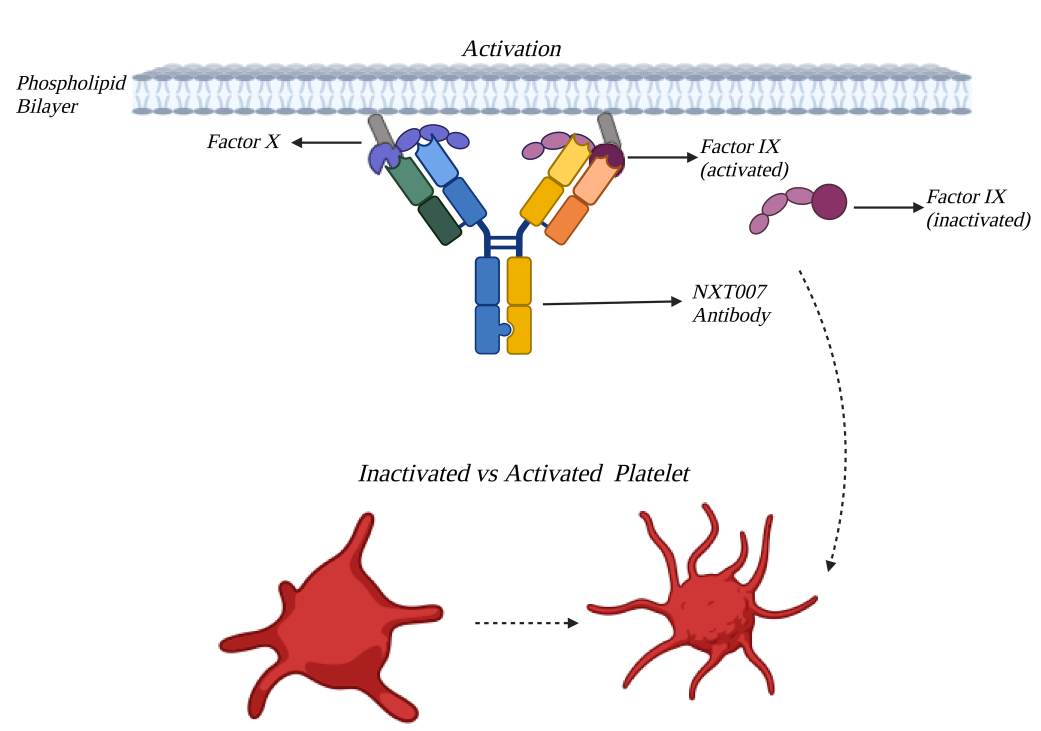
INTRODUCTION



- To help Hemophilia A patients, a drug called emicizumab which helps bridge the gap between Factor IXa and Factor X, taking on the role of Factor VIII to make clotting faster.
- Although emicizumab is a huge improvement for Hemophilia A patients, it is still not as good as the real Factor VIII, meaning there is a lot of room for improvement in this antibody.

METHODS

We have made a new antibody called NXT007, which is much more effective than emicizumab, and does its work with much more efficiency, accuracy, and sustainability.



NXT007's special characteristic is its unique binding arm shapes, which are not bivalently identical, unlike the majority of antibodies. This difference allows NXT007 to efficiently bind to both Factors IXa and X through binding arms individually engineered to do so.

RESULTS

- Our tests with NXT007 proved that it is 21x to 23x more potent than emicizumab.
- Probably the most important result, NXT007 achieved clotting activity which was extremely close to normal factor VIII levels. It reached activity comparable to about 100 IU/dL of Factor VIII, while emicizumab reached only like 40 IU/dL. For context, Factor VIII's normal levels are around 50-150 IU/dL.
- Furthermore, we measured that NXT007 has about 20x greater antifibrinolytic potency than emicizumab.
- Lastly, a lot less NXT007 was required (~30-100 μg/mL) to reach peak clotting than emicizumab (~100-300 μg/mL).

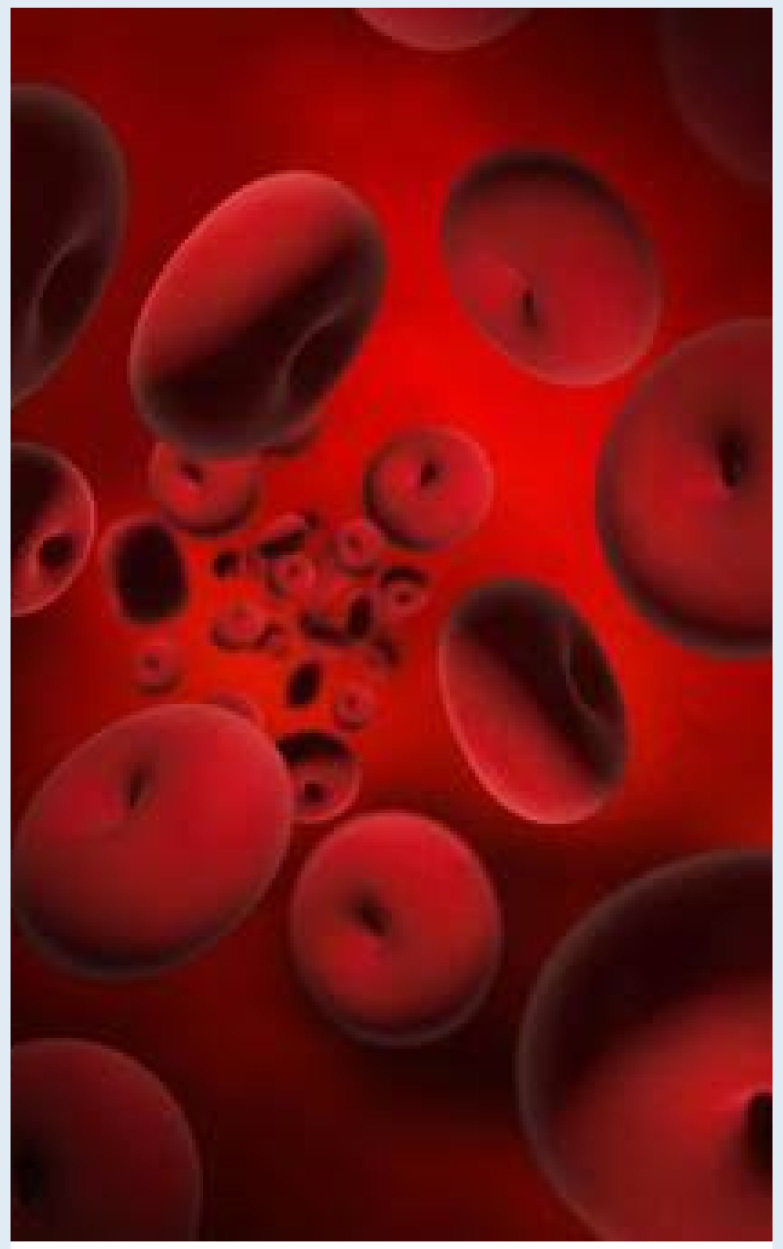
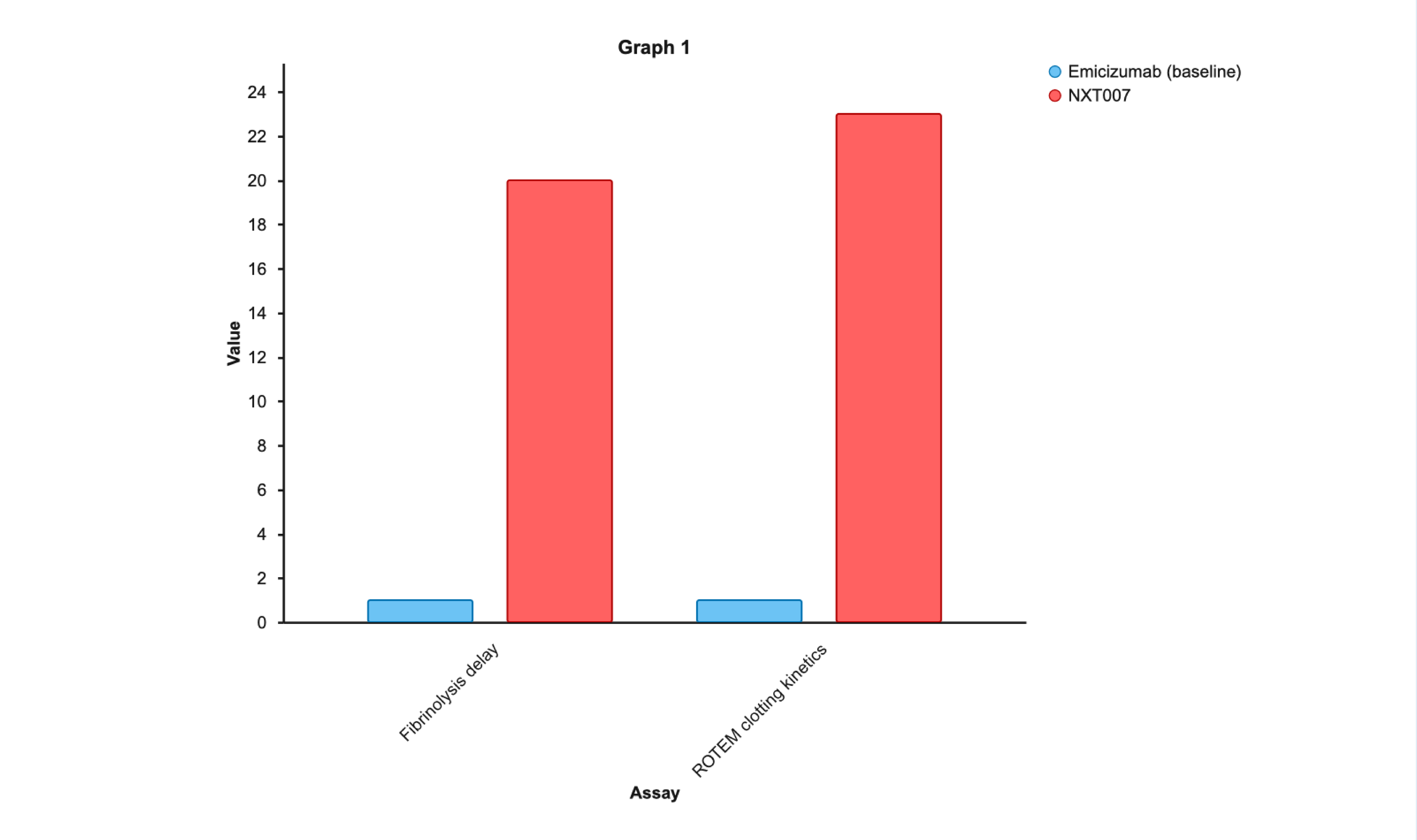


Illustration of red blood cells. Image courtesy of PhotoAC.



CONCLUSION

Currently, NXT007 is harder to produce than emicizumab due to the unique mechanism of each binding arm of the antibody, but we are researching ways to make production easier and cheaper. We believe that it can help Hemophilia A patients live a normal life, free from all the pain and anxiety that comes with such a debilitating disease.

CITATIONS

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